

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Geology and Geophysics

2. **Subject Name:** Seismic Methods of Prospecting

L-T-P: 3-1-0

Credits:4

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus: *An overview of Hydrocarbon accumulations and its surface exploration. A review of basic seismic principles: Fermat's Principle. Huygen's Principle. Reciprocity. Generalized Snell's law. Seismic Events: Reflection, Refraction, Multiple and Diffraction from multilayered media. Rayleigh criterion and Fresnel zone. Geometry of Seismic Wavepaths: Reflection Paths in constant velocity layer and in multi-layered earth model, Refraction Paths. Seismic Velocity, Group & Phase Velocities, Acoustic Impedance. Reflection and Transmission coefficients – Zoeppritz's equation. Propagation model for exploration seismology. Relationship between geological-physical causes, seismic facies and interpretation.*

Seismic Data Acquisition: Seismic Energy Sources: Dynamite, Vibroseis, Airgun etc. Seismic Detectors: Geophone, Hydrophone, Marshphone, Geophone string. Seismic noises and Noise Profile Analysis. Source and detector arrays. Directional shooting. Shooting geometry and Conventional Seismic survey design. Instrumentation in Refraction & Reflection surveys. Digital seismic data recording: General constitution, Analog module, Controller module and Tape transport module, pre-amplifier filters, filters, multiplexer, sample-and-hold, A-D converter, AGC, IFP, the recorder; SEG tape formats :SEGB, SEGC, SEGD and SEG Y. Multichannel Telemetry systems: both wire-line and radio telemetry.

Refraction Seismics: Correction to a datum. Interpretation of refraction data by graphical (Mean-Minus-T & Hale's and wavefront methods) and analytical (Generalized reciprocal method). Refraction Tomography.

Reflection Seismics: Common Depth Point Seismic Reflection Shooting (CDP). 2-D Seismic Reflection Acquisition parameters and survey design. 3-D Seismic Reflection Acquisition Parameters and survey design. Stacking chart and CDP/CMP gather diagram. Stacking Foldage (2-D shooting), In-line and Cross-line Foldage (3-D shooting). CDP bin. High Resolution Seismic Data Acquisition. An overview of 2-D CDP reflection Seismic Data Processing. Demultiplex. Exp-gain compensation. Mute. Edit. Field geometry to convert DE multiplexed shot sequential data to CDP/CMP gathers. Prestack partial migration (dip moveout). Space and time varying filtering. Spiking/Signature Deconvolution. Field Static corrections and surface consistent residual static corrections. Velocity analysis by Spectrum, Constant Velocity Gather, Constant Velocity Stack and Horizon Velocity Analysis. Normal Moveout (NMO) correction. Choice of coherency measures. Predictive Deconvolution. Space and Time Varying Filtering. Time Variant Spectral Whitening technique (optional). F-K filtering (optional). Stack. Radon Transform and slant stack. Migration after stack (Kirchhoff,

Finite Difference and Frequency-wavenumber migration- Gazdag / Stolt schemes). Depth Migration. Spherical divergence and correction for geometric spreading. Wavelet Processing (optional). Overview of 3-D Seismic Data Processing. Attribute analysis. Geological interpretation of seismic reflection stack and migrated CDP/CMP data. Amplitude Versus Offset (AVO) analysis and Seismic Reservoir Characterization, 4-D Seismic Technology and Time-lapse Seismics.

Vertical Seismic Profiling: Fundamentals of VSP method, Geogram, Check-shot survey. Separation of up-going and down-going waves and VSP data processing. Corridor stack.

Space for entering the syllabus

Reference Books:

- 1) Introduction to Geophysical Prospecting by Milton B. Dobrin and Carl H. Savit
- 2) Applied Geophysics by W. M. Telford, L. P. Geldart and R. E. Sheriff
- 3) Applied Seismology by Mamdouh R. Gadallah and Ray L. Fisher
- 4) Seismic Data Processing Volume I and II by O. Yilmaz
- 5) Introduction to Petroleum Seismology by Luc T. Ikelle and Lasse Amundsen

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	An overview of Hydrocarbon accumulations and its surface exploration. A review of basic seismic principles	08
2.	Seismic Data Acquisition	08
3.	Refraction Seismics	04
4.	Reflection Seismics	16
5.	Amplitude Versus Offset (AVO) analysis and Seismic Reservoir Characterization	06
6.	4-D Seismic Technology and Time-lapse Seismics.	04
7.	Vertical Seismic Profiling	02
Total number of hours		48